GOLDEN BOT
COMPLETE TRAINING GUIDE

This document explains the complete training ecosystem of Golden Bot and Brain Trader Pro style AI trading systems. Topics include: - Primary model training - Ensemble learning - Triple barrier labeling - Meta-labeling - Calibration - Drift detection - Shadow challenger models - Auto-retraining - Training commands - Hyperparameter tuning - Good vs bad practices - Professional quant workflows

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1. PRIMARY MODEL TRAINING

The primary model predicts trade direction and probability of profitable outcomes.

Core models:
- XGBoost
- LightGBM

The ensemble combines both predictions.

Training flow:
1. Download historical Binance data
2. Build features
3. Apply triple barrier labels
4. Train XGBoost
5. Train LightGBM
6. Average predictions
7. Save model

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2. PRIMARY TRAINING COMMAND

Windows:

```
python train_primary.py ^
--symbols BTCUSDT,ETHUSDT,SOLUSDT,BNBUSDT,XRPUSDT ^
--years 3 ^
--market futures ^
--interval 15m ^
--output models/ensemble_top5.pkl
```

Linux/macOS:

```
python train_primary.py \
--symbols BTCUSDT,ETHUSDT,SOLUSDT,BNBUSDT,XRPUSDT \
```

--years 3 \
--market futures \
--interval 15m \
--output models/ensemble_top5.pkl

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3. TRAINING PARAMETERS

--symbols
Coins used for training.

Recommended:
BTCUSDT, ETHUSDT, SOLUSDT, BNBUSDT, XRPUSDT

--years
Historical years.

Recommended:
2–4 years

--market
futures or spot

Recommended:
futures

--interval
Candle timeframe.

Recommended:
15m

5m:
More aggressive, more noise.

1h:
Cleaner but slower.

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4. TRIPLE BARRIER LABELING

Instead of simple future-return labels, the bot uses:

- Take profit barrier

- Stop loss barrier
- Time expiration barrier

This creates realistic trade labels.

Regime-aware settings:

TREND:
TP = 2.5 ATR
SL = 1.2 ATR

CHOP:
TP = 1.0 ATR
SL = 0.8 ATR

HIGHVOL:
TP = 3.0 ATR
SL = 2.0 ATR

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5. FEATURE ENGINEERING

Feature engineering converts market candles into machine-learning inputs.

Features include:
- RSI
- MACD
- ATR
- ADX
- Momentum
- Volume spikes
- Volatility
- VWAP
- Trend strength

Good features:
- stable
- normalized
- regime-aware

Bad features:
- noisy
- unstable
- heavily correlated

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6. XGBOOST TRAINING

Typical parameters:

n_estimators=300
max_depth=6
learning_rate=0.05
subsample=0.8
colsample_bytree=0.8

Good settings:
- lower learning rate
- moderate depth
- balanced subsampling

Bad settings:
- huge depth
- aggressive learning rate
- too many estimators

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7. LIGHTGBM TRAINING

LightGBM provides a second learning architecture.

Purpose:
- diversity
- robustness
- reduced overfitting

Final prediction:

(XGBoost + LightGBM) / 2

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8. META-LABELER TRAINING

Meta-labeling filters weak trades.

The primary model says:
"Trade looks good."

The meta-model asks:
"Will this still be profitable after fees and slippage?"

Meta features:
- primary confidence
- ADX
- BTC returns
- volume spikes
- ATR percentile
- regime ID

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9. META TRAINING COMMAND

Windows:

```
python train_meta.py ^
--primary-model models/ensemble_top5.pkl ^
--symbols BTCUSDT,ETHUSDT,SOLUSDT,BNBUSDT,XRPUSDT ^
--years 2 ^
--fee 0.001 ^
--output models/meta_top5.pkl
```

Linux/macOS:

```
python train_meta.py \
--primary-model models/ensemble_top5.pkl \
--symbols BTCUSDT,ETHUSDT,SOLUSDT,BNBUSDT,XRPUSDT \
--years 2 \
--fee 0.001 \
--output models/meta_top5.pkl
```

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10. PROBABILITY CALIBRATION

Raw probabilities are often unrealistic.

Calibration methods:
- isotonic regression
- Platt scaling

Benefits:
- realistic confidence
- improved risk management
- better position sizing

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11. DRIFT DETECTION

Markets evolve continuously.

Drift detection identifies:
- changing feature distributions
- model degradation
- unstable predictions

Responses:
- retrain model
- lower confidence
- reduce risk
- activate challenger model

12. SHADOW CHALLENGER MODELS

New models run in paper-trading mode.

Compared against production model.

Promotion conditions:
- higher Sharpe
- lower drawdown
- better expectancy
- more stable returns

13. AUTO-RETRAINING

Retraining triggers:
- weekly schedule
- drift detection
- performance degradation

Recommended:
Futures → weekly
Spot → monthly

14. VALIDATION METHODS

Correct validation:
Old data → train
New data → test

Avoid random shuffling because it leaks future information.

15. WALK-FORWARD VALIDATION

Train:
2022–2024

Test:
2025

Then roll forward.

This simulates real trading conditions.

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16. GOOD VS BAD PRACTICES

GOOD:
- realistic fees
- realistic slippage
- liquid coins
- weekly retraining
- meta filtering

BAD:
- meme coins
- overfitting
- unrealistic backtests
- excessive leverage
- random train/test splits

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17. RECOMMENDED BEGINNER SETUP

Coins:
BTC ETH SOL BNB XRP

Years:
3

Market:
futures

Interval:
15m

This setup offers:
- stable liquidity
- manageable noise
- balanced trade frequency

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18. ADVANCED TRAINING IDEAS

Advanced systems may include:
- regime-specialized models
- volatility models

- cross-sectional ranking
- online learning
- adaptive thresholds
- reinforcement layers

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19. PROFESSIONAL QUANT STACK

Professional architecture:
1. Ensemble models
2. Triple barrier labels
3. Meta-labeling
4. Calibration
5. Drift detection
6. Challenger systems
7. Walk-forward validation
8. Risk-aware execution
9. Auto-retraining

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20. FINAL ADVICE

The strongest trading systems usually win because of:
- filtering bad trades
- consistency
- risk management
- adaptation
- avoiding overfitting

A stable 55–60% edge with proper risk management is more powerful than unstable high-win-rate systems.
